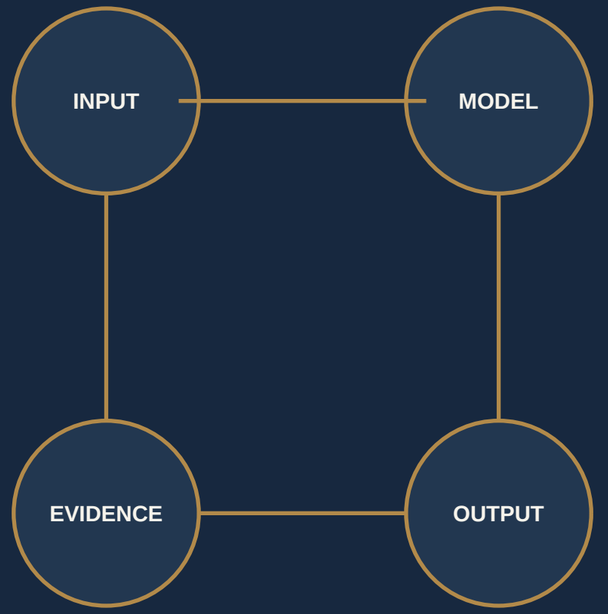
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Fuzzy Logic

AI.209 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define fuzzy sets.
- Apply key methods including membership functions.
- Build or analyze a practical system involving fuzzy rules.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Fuzzy sets

02

METHODS

Membership functions

03

SYSTEMS

Fuzzy rules

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Fuzzy sets

Fuzzy sets

02 · Membership functions

Membership functions

03 · Fuzzy rules

Fuzzy rules

04 · Inference

Inference

05 · Defuzzification and fuzzy cont

Defuzzification and fuzzy controllers

06 · Mandatory practical: Build fu

Mandatory practical: Build fuzzy controllers for temperature

07 · Motor speed and robotic system

Motor speed and robotic systems

Core ideas, made explicit

AI.209 • Distinguish definitions, assumptions and evidence

- Fuzzy sets
- Membership functions
- Fuzzy rules
- Inference
- Defuzzification and fuzzy controllers

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.209 • Distinguish definitions, assumptions and evidence

- Mandatory practical: Build fuzzy controllers for temperature
- Motor speed and robotic systems

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving fuzzy sets.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using fuzzy sets.
- Implement or simulate a baseline using membership functions.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.